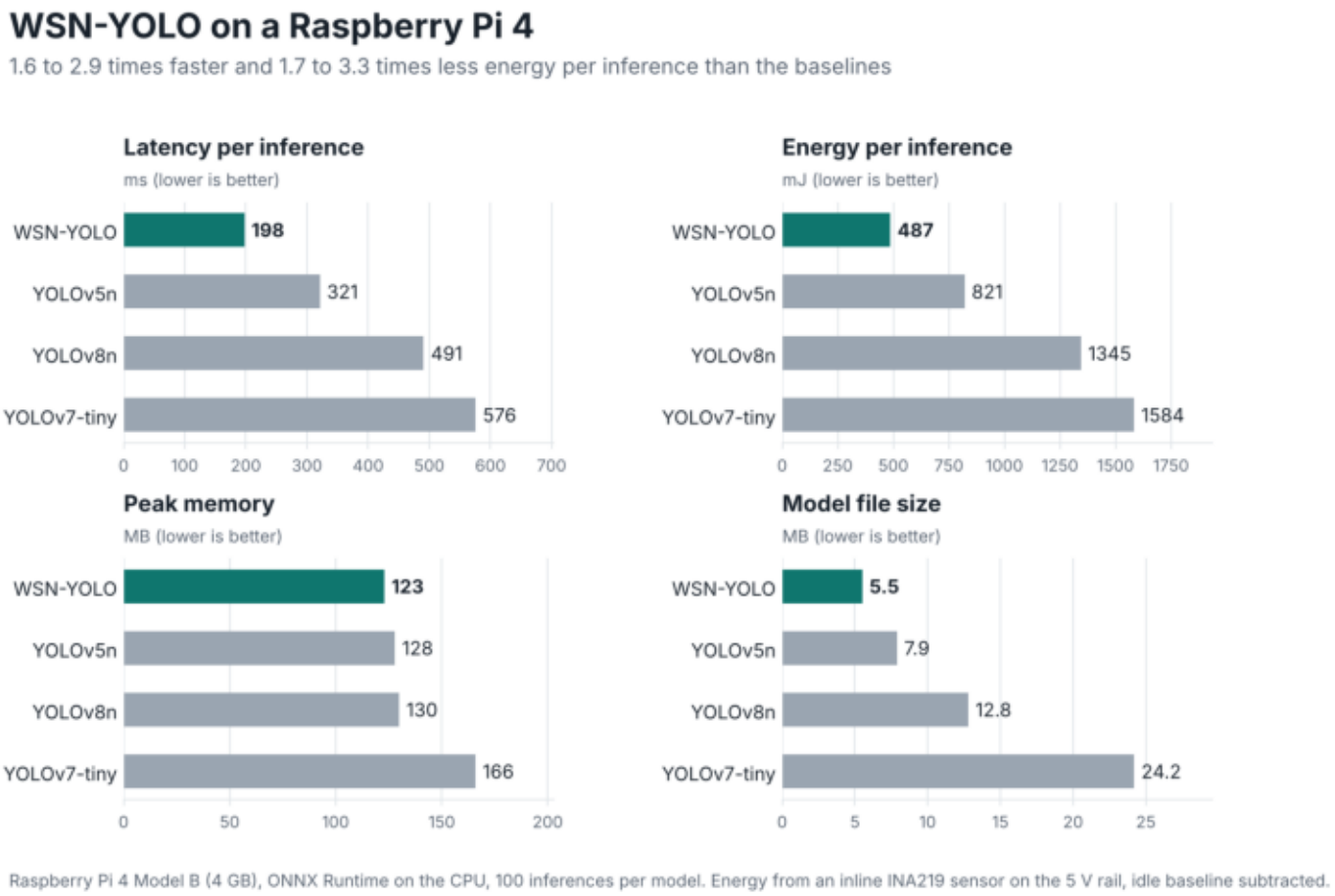


Summary

WSN-YOLO runs at 197.6 ms per inference on a Raspberry Pi 4, in a 5.5 MB model that uses 123 MB of memory.

Each inference costs 487 mJ at 4.77 W. YOLOv5n needs 821 mJ, YOLOv8n 1345 mJ and YOLOv7-tiny 1584 mJ.

Power draw is similar for all four models (4.8 to 5.2 W), so the energy gap comes mainly from how long each model keeps the processor busy.

Method

Raspberry Pi 4 Model B (4 GB), ONNX Runtime on the CPU, 100 inferences per model. Energy from an inline INA219 sensor on the 5 V rail, idle baseline subtracted.

Latency is the mean over the runs, with the 95th percentile within 2 ms of it for every model. Numbers are in pi_results.csv.